SAIence NSX-1

NSX-1 Scene Recall Help

Inputs, Output, Parameter and Command of the SIMPL+ Module

Item	Description
Product	NSX-1 lighting control system (lighting modules connected by RS485)
Software	SIMPL+ module `NSX-1 Scene Recall.usp` (3-Series and 4-Series) SIMPL Windows
Audience	Programmers of Crestron control processors
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1. Scope of This Document

This document describes the inputs, output and parameter of the SIMPL+ module NSX-1 Scene Recall, and the command it sends. The module receives digital signals such as touch panel buttons, builds the command that recalls a scene for an NSX-1 group, and outputs it to a COM port.

- Names and functions of the symbol
- Format of the command
- Use in a SIMPL Windows program
- Reference Name and the Log$ output

Group and scene settings of the NSX-1 lighting modules, and v commands (on, off, dimming), are not covered.

1.1 Terms

Term	Meaning
Command	Character string sent on the NSX-1 line. This module sends a 12-character scene recall command.
Group number	Number of a set of lighting circuits (1 to 999). 3 digits in the command.
Scene number	Number of a combination of light levels for a group. 4 digits in the command. The levels are stored in the lighting modules.
Reference Name	Field that SIMPL Windows adds to every SIMPL+ symbol. A name can be given to each placed symbol.
Log$	Optional serial output of this module that shows `<Reference Name>: <command>` in SIMPL Debugger.

2. Names and Functions of the Symbol

Figure 2-1 shows the symbol of this module placed in a SIMPL Windows program.

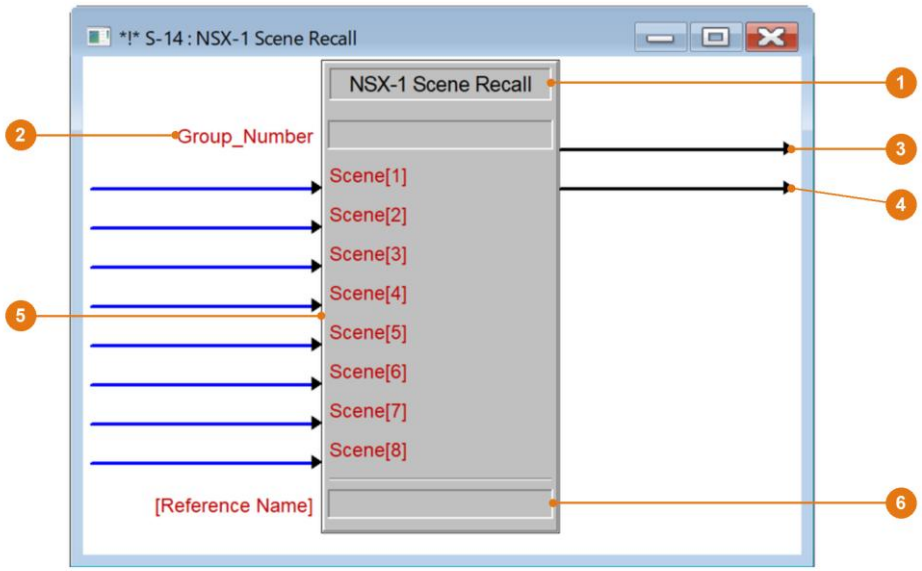


Figure 2-1 NSX-1 Scene Recall symbol

No.	Name	Type	Function
1	NSX-1 Scene Recall	Symbol name	Module name set by `#SYMBOL_NAME`.
2	Group_Number	Parameter (integer)	Group number placed in the command. Enter 1 to 999 as a decimal value such as `10d`. There is no default value, so the field shows the parameter name until a value is entered.
3	TX$	Output (serial)	The built command. Connect it to `TX$` of the COM port symbol.
4	Log$	Output (serial)	`<Reference Name>: <command>` for checking in SIMPL Debugger. May be left unconnected (Chapter 5).
5	Scene[1] to Scene[8]	Inputs (digital)	A rising edge (press) sends the command once. Scene[1] is scene 0 and Scene[8] is scene 7.
6	Reference Name	SIMPL Windows field	Name of the symbol, placed at the start of Log$. It may be left blank (Chapter 5).

Note

SIMPL Windows draws the parameter field on the same row as the first input. The module declares `DIGITAL_INPUT _SKIP_;` before the inputs to keep the first row empty, so the Group_Number field does not cover Scene[1].

3. Command Sent by the Module

3.1 Command Format

The command has the following 12 characters. No terminator such as CR or LF is added.

Position	Characters	Content
1	`g`	Indicates a command addressed to a group
2 to 4	`010`	Group number (3 digits, zero-padded). Value of Group_Number
5 to 8	`sccc`	Scene recall instruction
9 to 12	`0001`	Scene number (4 digits, zero-padded). Number of the pressed input minus 1

The command is built by the SIMPL+ function `MakeString` with the format `"g%03dsccc%04d"`.

3.2 Command Examples

Group_Number	Pressed input	Command sent	Result
10d	Scene[1]	`g010sccc0000`	Group 010 recalls scene 0
10d	Scene[2]	`g010sccc0001`	Group 010 recalls scene 1
10d	Scene[3]	`g010sccc0002`	Group 010 recalls scene 2
20d	Scene[8]	`g020sccc0007`	Group 020 recalls scene 7

> **Note**
> The lighting modules do not reply to a scene command, and this module does not wait for a reply.

4. Use in a SIMPL Windows Program

4.1 Preparing the Module

1. Put `NSX-1 Scene Recall.usp` and `NSX-1 Scene Recall.pdf` in the user SIMPL+ folder of SIMPL Windows.
2. Open `NSX-1 Scene Recall.usp` in SIMPL+, select 3-Series and 4-Series in Build > Control System Target, and compile.
3. Check that the time stamp of `NSX-1 Scene Recall.ush` has been updated. The module appears under User SIMPL+ in the SIMPL Windows symbol library.

4.2 Placing and Connecting the Symbol

1. Place NSX-1 Scene Recall in the SIMPL Windows program, one symbol per group.
2. Enter the group number in the Group_Number field (for example `10d`).
3. Connect digital signals such as touch panel buttons to Scene[1] to Scene[8]. Unused inputs may be left unconnected.
4. Connect TX$ to `TX$` of the COM port connected to the NSX-1 line. The TX$ outputs of several symbols may use the same signal name.
5. If required, enter a name such as a room name in Reference Name, and give Log$ a signal name to check it in SIMPL Debugger (Chapter 5).

4.3 COM Port Settings

Item	Setting
Protocol	RS485
Baud rate	9600 bps
Data bits, parity, stop bits	8, none, 1
Wiring (example: COM1 of CP4)	RTS = + (TX+/RX+), TX = - (TX-/RX-), G = GND. RX and CTS are not used.

5. Reference Name and the Log$ Output

Reference Name is a field that SIMPL Windows adds below every SIMPL+ symbol. A name can be given to each placed symbol, and SIMPL+ code reads it with `GetSymbolReferenceName()`. Each time this module sends a command, it outputs the name and the command to Log$.

```
MakeString(Log$, "%s: %s", GetSymbolReferenceName(), sCmd);
```

Reference Name	Pressed input (Group_Number = 10d)	Value of Log$
Living	Scene[2]	`Living: g010sccc0001`
(blank)	Scene[2]	`: g010sccc0001`

Give Log$ a signal name (for example `NSX_LOG`) and watch it in SIMPL Debugger. When several instances of the module are placed, Log$ shows which symbol sent each command. A blank Reference Name, or an unconnected Log$, does not change the command or the operation.

6. Notes and Known Issues

Item	Description
Compile target	For 3-Series and 4-Series. X-Generation (2-Series) does not support `INTEGER_PARAMETER` and similar declarations, so compiling for it fails.
Group_Number	There is no default value. Enter a value for each placed symbol. Range: 1 to 999.
Scene number range	One symbol recalls scenes 0 to 7. For other scene numbers, change `MAX_SCENE` or send the command by another method.
Symbol rows	After changing the number of inputs or parameters in SIMPL+, compile and place the symbol again in SIMPL Windows. An existing instance keeps the old row positions.
Line endings of the source	Save the `.usp` file with CRLF line endings. A file with LF only fails to compile (Line 1 Missing }).
SIMPL+ Trace	Messages from the SIMPL+ `Trace` function did not appear in SIMPL Debugger or the Text Console in a test on site. This module therefore uses the Log$ output instead.
Reply	The lighting modules do not reply to a scene command. Confirm the result by the state of the lights.

Appendix A SIMPL+ Code

Full text of `NSX-1 Scene Recall.usp`.

```
/*******************************************************************************************
  SIMPL+ Module Information
*******************************************************************************************/
/*
Dealer Name:   SAIence
System Name:   NSX-1
Programmer:
Comments:      Builds the NSX-1 scene recall command g(3-digit group)sccc(4-digit scene number) and sends
it to TX$
               Scene[1] = scene 0, Scene[2] = scene 1 ... Scene[8] = scene 7
               Connect TX$ to the TX$ of the COM port (RS485, 9600 bps, 8N1)
*/


/*******************************************************************************************
  Compiler Directives
*******************************************************************************************/
#SYMBOL_NAME "NSX-1 Scene Recall"
#HELP_PDF_FILE "NSX-1 Scene Recall.pdf"
#DEFINE_CONSTANT MAX_SCENE 8


#DEFAULT_VOLATILE
#ENABLE_STACK_CHECKING
#ENABLE_TRACE


/*******************************************************************************************
  DIGITAL, ANALOG and SERIAL INPUTS and OUTPUTS
*******************************************************************************************/
DIGITAL_INPUT  _SKIP_;              // row 1 is used by the Group_Number parameter field
DIGITAL_INPUT  Scene[MAX_SCENE];
STRING_OUTPUT  TX$;
STRING_OUTPUT  Log$;                // "<Reference Name>: <command>" for SIMPL Debugger (optional)


/*******************************************************************************************
  Parameters
*******************************************************************************************/
INTEGER_PARAMETER Group_Number;


#BEGIN_PARAMETER_PROPERTIES Group_Number
    propValidUnits = unitDecimal;
    propBounds = 1d , 999d;
    // no propDefaultValue: until a value is entered, SIMPL Windows shows the name Group_Number in this
field
    propShortDescription = "NSX-1 group number (1-999)";
#END_PARAMETER_PROPERTIES


/*******************************************************************************************
  Event Handlers
*******************************************************************************************/
PUSH Scene
{
    INTEGER nScene;
    STRING sCmd[20];
    nScene = GetLastModifiedArrayIndex() - 1;   // Scene[1] = scene 0
    MakeString(sCmd, "g%03dsccc%04d", Group_Number, nScene);
    TX$ = sCmd;
    MakeString(Log$, "%s: %s", GetSymbolReferenceName(), sCmd);   // Reference Name: sent command
}


/*******************************************************************************************
  Main()
*******************************************************************************************/
Function Main()
{
    WaitForInitializationComplete();
}
```